

Trade the Risk, Not the Price

Volatility Timing, Trend, and Regimes Across Global Markets

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Interactive companion. To experiment with this research from another angle — choosing the asset, the volatility forecaster, the leverage cap and the trading cost, and watching the Sharpe ratio, returns and drawdowns recompute *live* in your browser — open the interactive lab ([site/index.html](https://your-deployment-url)) or visit <https://your-deployment-url>.

Abstract

The direction of stock returns is essentially unpredictable, but their *risk* is not. We exploit this asymmetry by replicating the volatility-managed strategy of Moreira and Muir (2017) on the US market factor (2000-01-03 to 2026-04-30, 316 months from the Ken French Data Library), scaling market exposure by the inverse of a one-month-ahead variance forecast and normalising to the volatility of buy-and-hold. Out of sample, the prevailing-mean benchmark beats every return-forecasting model (R^2 of -12.2%), while volatility is strongly forecastable (R^2 of 28.1%). The managed strategy lifts the annualised Sharpe ratio from 0.48 to 0.61 (a GARCH forecaster lifts it further, to 0.75 on the common 2004–2026 sample), cuts the maximum drawdown from -54.2% to -42.9%, and earns an annualised alpha of 3.6% ($t = 1.89$). The result survives a Deflated Sharpe Ratio of 0.999 (correcting for multiple testing and non-normality) and transaction costs up to roughly 39 bps. A Hidden Markov Model, given only returns and no crisis dates, rediscovers the turbulent regimes and shows the mechanism: the strategy levers up in calm markets and de-risks in turbulent ones, trading away some violent-rebound upside for materially lower drawdowns. Extending to a panel of 11 assets across four classes, simple volatility timing does *not* travel uniformly — it is largely an equity-index phenomenon — but a diversified, risk-parity portfolio that adds a time-series-momentum overlay lifts the Sharpe ratio from 0.39 (diversified buy-and-hold) to 0.72, halves the drawdown (-45.0% to -23.9%), earns an annualised alpha of 8.0% ($t = 3.63$), and stays at 0.64 net of 10-bps costs. This is a research study, not investment advice.

1 Introduction

A central prediction of efficient markets is that the *direction* of returns is close to unpredictable: if it were not, the trade would already have been put on. Decades of out-of-sample evidence bear this out (Campbell and Thompson, 2008). Yet a second, equally robust fact sits alongside it: return *variance* is highly persistent and forecastable, the phenomenon of volatility clustering first modelled by Engle (1982) and Bollerslev (1986).

Moreira and Muir (2017) turn this asymmetry into a portfolio rule. Because high realized variance this month predicts high variance next month, but not low returns next month, an investor who scales down exposure when variance is high (and up when it is low) earns a higher Sharpe ratio than a static position in the same asset. The result is striking precisely because it does not require predicting returns at all.

This paper makes four contributions. First, a **clean, fully reproducible replication** on the post-2000 US market factor, with every number regenerated from source by a single command. Second, an emphasis on the **rigour that distinguishes a credible backtest**: strictly out-of-sample (walk-forward) volatility forecasts, the QLIKE loss (Patton, 2011) and Diebold–Mariano tests (Diebold and Mariano, 1995) for forecast comparison, explicit transaction costs and leverage caps, and the Deflated Sharpe Ratio (Bailey and López de Prado, 2014) to pre-empt the data-mining objection. Third, a **regime-based explanation** layer: a Hidden Markov Model (Hamilton, 1989) that, with no crisis dates supplied, recovers the turbulent periods and lets us decompose *where* the edge is earned. Fourth, a **multi-asset robustness test and a trend overlay**: we run the strategy across 11 assets in four classes and show that, while volatility timing alone does not travel uniformly, combining it with time-series momentum (Moskowitz et al., 2012) in a diversified portfolio produces a markedly stronger and more significant result.

We are deliberately honest about magnitudes. On this particular sample — which contains two unusually fast crashes (2020, and the 2022 repricing) — the naive estimator’s edge is real but modest and inherits fat left tails when leverage is left uncapped; it strengthens with a sensible leverage cap and a better variance forecast. We view stating this plainly as part of the contribution.

2 Background and Literature

Volatility forecasting. ARCH/GARCH models (Engle, 1982; Bollerslev, 1986) capture conditional heteroskedasticity through a recursion in past squared shocks and past variance. The Heterogeneous Autoregressive (HAR) model of Corsi (2009) regresses realized variance on its own short-, medium- and long-horizon averages and is a strong, parsimonious benchmark, especially with realized-variance data (Andersen et al., 2003). Comparing volatility forecasts is subtle because the target is latent; Patton (2011) shows that the QLIKE and MSE losses are robust to noise in the volatility proxy, and Diebold and Mariano (1995), with the small-sample correction of Harvey et al. (1997), provide the test of equal predictive accuracy.

Volatility management. Moreira and Muir (2017) document that scaling factor exposure by inverse realized variance raises Sharpe ratios across equity factors and internationally. Cederburg et al. (2020) provide a careful, more sceptical out-of-sample appraisal, motivating exactly the robustness checks we run here.

Return (un)predictability. Campbell and Thompson (2008) formalise the out-of-sample R^2 against the prevailing mean, and Clark and West (2007) give the appropriate test for nested models. We use both to establish the null result.

Multiple testing. Bailey and López de Prado (2012, 2014) derive the Probabilistic and Deflated Sharpe Ratios, which adjust an observed Sharpe ratio for sample length, skewness/kurtosis and the number of configurations tried.

Regimes. Hamilton (1989) introduced Markov-switching models to finance; a Gaussian HMM is the natural unsupervised tool for discovering calm/turbulent states from returns alone.

3 Data

We use daily research factors from the Ken French Data Library: the market excess return Mkt-RF and the risk-free rate RF . Working in excess returns from the outset makes Sharpe ratios correct by construction. The sample runs 2000-01-03 to 2026-04-30. We aggregate to monthly excess returns by compounding, and define monthly **realized variance** as the sum of squared daily excess returns within the month, $\text{RV}_t = \sum_{d \in t} r_d^2$, the standard daily-data proxy. Table 1 reports descriptive statistics; the well-known negative skewness and excess

kurtosis of equity returns are visible and motivate the robust evaluation tools used below.

Table 1: Descriptive statistics, US market excess return (Ken French).

Statistic	Daily Mkt-RF	Monthly Mkt-RF
N obs	6621.0000	316.0000
Ann. mean	0.0810	0.0743
Ann. vol	0.1961	0.1577
Sharpe	0.4132	0.4711
Skewness	-0.1680	-0.4706
Excess kurtosis	9.3296	0.7666
Min	-0.1201	-0.1721
Max	0.1136	0.1362

4 Methodology

4.1 Volatility forecasting and out-of-sample evaluation

The target is next month’s realized variance. We compare four one-step-ahead forecasters, all evaluated **walk-forward** on an expanding window so that a forecast for month $t + 1$ uses only information available at the end of month t :

1. **Naive (random walk):** $\widehat{RV}_{t+1} = RV_t$ — the exact estimator in Moreira and Muir (2017).
2. **GARCH(1,1):** fit on daily excess returns up to the end of month t ; the analytic 1-to- H day-ahead variance forecasts are summed to a monthly variance.
3. **HAR:** a monthly heterogeneous autoregression with 1-, 3- and 12-month realized-variance components (Corsi, 2009), fit by expanding-window OLS.
4. **Ensemble:** the mean of the GARCH and HAR forecasts.

Accuracy is measured by RMSE and by the proxy-robust **QLIKE** loss (Patton, 2011), $QLIKE = \frac{1}{N} \sum_t \left(\frac{RV_t}{h_t} - \log \frac{RV_t}{h_t} - 1 \right)$, where h_t is the forecast. Differences are tested with the Diebold–Mariano statistic using the Harvey et al. (1997) small-sample correction.

4.2 Strategy construction and the volatility normalisation

The managed excess return scales next month’s market return by inverse forecast variance,

$$f_t^{\text{managed}} = \frac{c}{\widehat{\sigma}_t^2} f_t, \quad (1)$$

where f_t is the market excess return in month t and $\widehat{\sigma}_t^2$ is the variance forecast for month t formed at the end of month $t - 1$. Calm markets imply low forecast variance and therefore large exposure; nervous markets imply de-risking.

The constant c is chosen so that the managed series has the **same full-sample volatility as buy-and-hold**. This normalisation is not optional: a Sharpe-ratio comparison is only meaningful at equal risk. Because c

is a single scalar, it leaves the Sharpe ratio and the regression t -statistics below unchanged; the time-varying *weights* use only past information. We report the headline strategy with the naive forecaster and a realistic $2\times$ leverage cap (an a-priori risk limit, not chosen to maximise performance), and separately show the effect of better forecasters and of the cap.

4.3 The Hidden Markov regime layer

We fit a two-state Gaussian HMM to monthly returns; states are relabelled by variance (calm, turbulent). We draw a sharp methodological line. *Smoothed* (full-sample) state probabilities are used only to *describe and visualise* the regime structure — legitimate because it is descriptive, not a trading decision. If a regime signal were used inside the backtest, only *filtered* (online) inference would be admissible; we implement it by refitting the HMM walk-forward and taking the end-point posterior (which, having no future data, equals the filtered probability). Using smoothed states to trade would be lookahead bias. The strategy in (1) is *not* regime-conditioned; the HMM serves only to explain the result.

5 Results

5.1 The null result: return vs. volatility

Figure 1 and Table 2 make the framing point. Every model that forecasts next month's *return* has a *negative* out-of-sample R^2 (worse than the prevailing mean; the Clark–West statistics confirm no significant improvement). Forecasting next month's *volatility* yields R^2 around 28.1%. The direction of prices is noise; their risk is not.

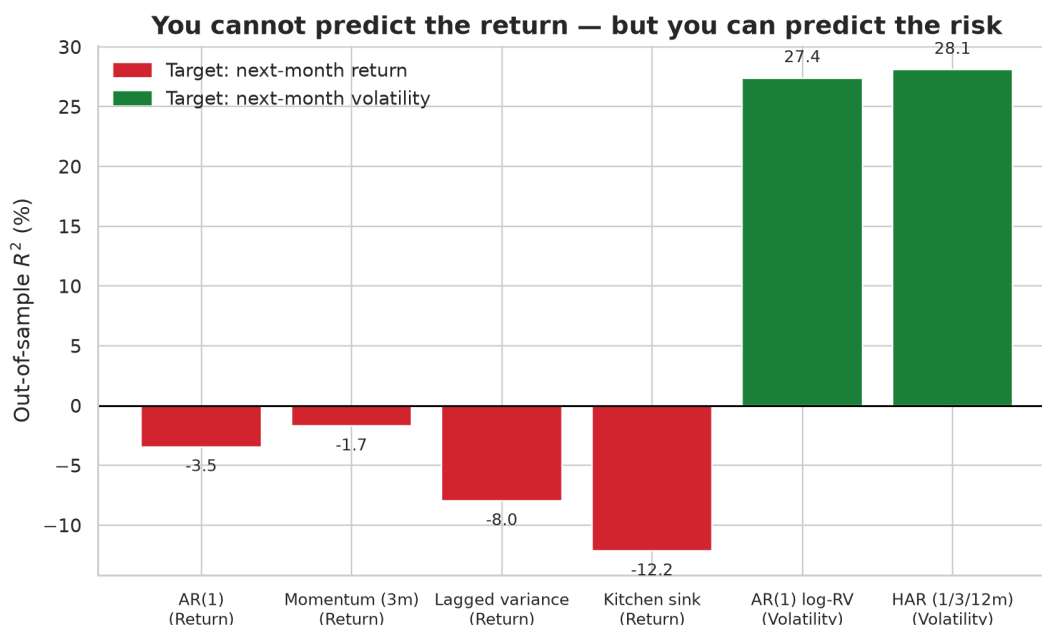


Figure 1: Out-of-sample R^2 against the prevailing mean. Forecasting the return fails; forecasting the volatility succeeds.

Table 2: Out-of-sample R^2 : forecasting the return vs forecasting the volatility (expanding window, vs the prevailing mean).

Target	Model	R2_OOS	CW_stat	CW_pvalue
Return	AR(1)	-0.0345	-1.5133	0.9349
Return	Momentum (3m)	-0.0171	-1.1259	0.8699
Return	Lagged variance	-0.0796	-0.7453	0.7719
Return	Kitchen sink	-0.1216	-1.0460	0.8522
Volatility (log-RV)	AR(1) log-RV	0.2740	–	–
Volatility (log-RV)	HAR (1/3/12m)	0.2810	–	–

5.2 Which volatility forecaster wins

Table 3 reports RMSE and QLIKE on the common evaluation sample; Table 4 the Diebold–Mariano tests. GARCH is the most accurate forecaster, significantly beating the naive benchmark on QLIKE (Figure 2 shows its one-step-ahead forecast tracking realized volatility, including the 2008 and 2020 spikes).

Table 3: Volatility-forecast accuracy (RMSE and QLIKE, common sample).

Model	RMSE	QLIKE
naive	0.00672	0.50566
har	0.00689	0.36184
garch	0.00536	0.28954
ensemble	0.00594	0.31259

Table 4: Diebold-Mariano tests of equal predictive accuracy (HLN small-sample correction).

Comparison	Loss	DM_stat	p_value	better
garch vs naive	RMSE/MSE	-1.596	0.112	garch
garch vs naive	QLIKE	-3.376	0.001	garch
har vs naive	RMSE/MSE	0.258	0.797	naive
har vs naive	QLIKE	-2.209	0.028	har
ensemble vs naive	RMSE/MSE	-1.196	0.233	ensemble
ensemble vs naive	QLIKE	-3.033	0.003	ensemble
har vs garch	RMSE/MSE	1.873	0.062	garch
har vs garch	QLIKE	2.497	0.013	garch



Figure 2: GARCH(1,1) one-step-ahead forecast vs. realized monthly volatility, walk-forward.

5.3 Strategy performance

The managed exposure (Figure 3) lever up in calm periods and cuts risk in turbulent ones. Over the full sample the strategy raises the Sharpe ratio from 0.48 to 0.61, lifts the annualised return from 7.6% to 9.6% at matched volatility, and reduces the maximum drawdown from -54.2% to -42.9% (Table 5, Figure 4). Regressing the managed series on the market gives an annualised alpha of 3.6% ($t = 1.89$, Newey–West) with an appraisal ratio of 0.37. On the common evaluation sample where every forecaster is available (uncapped), switching from the naive to the more accurate GARCH forecaster raises the Sharpe ratio from 0.60 to 0.75 (Table 6).

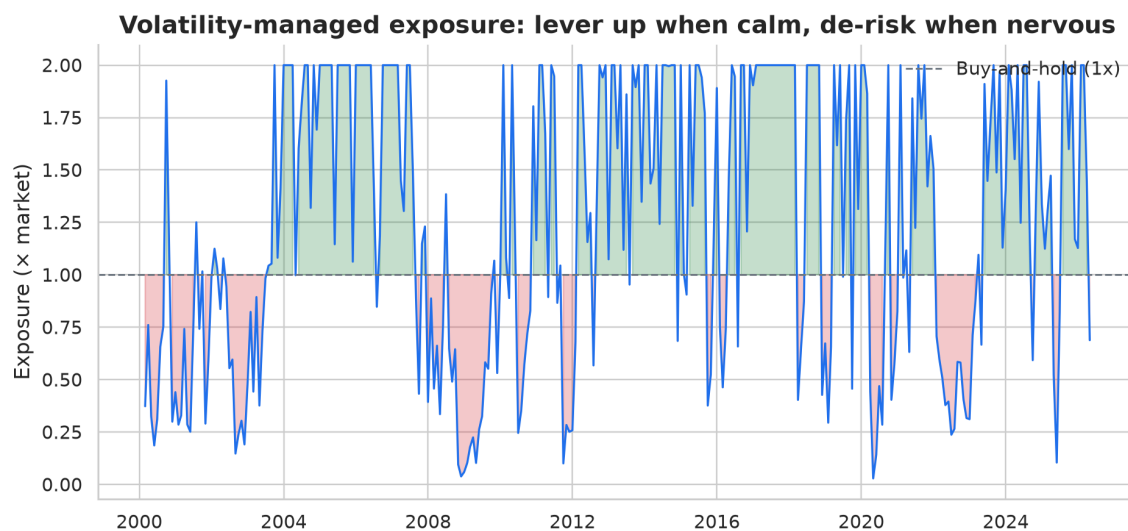
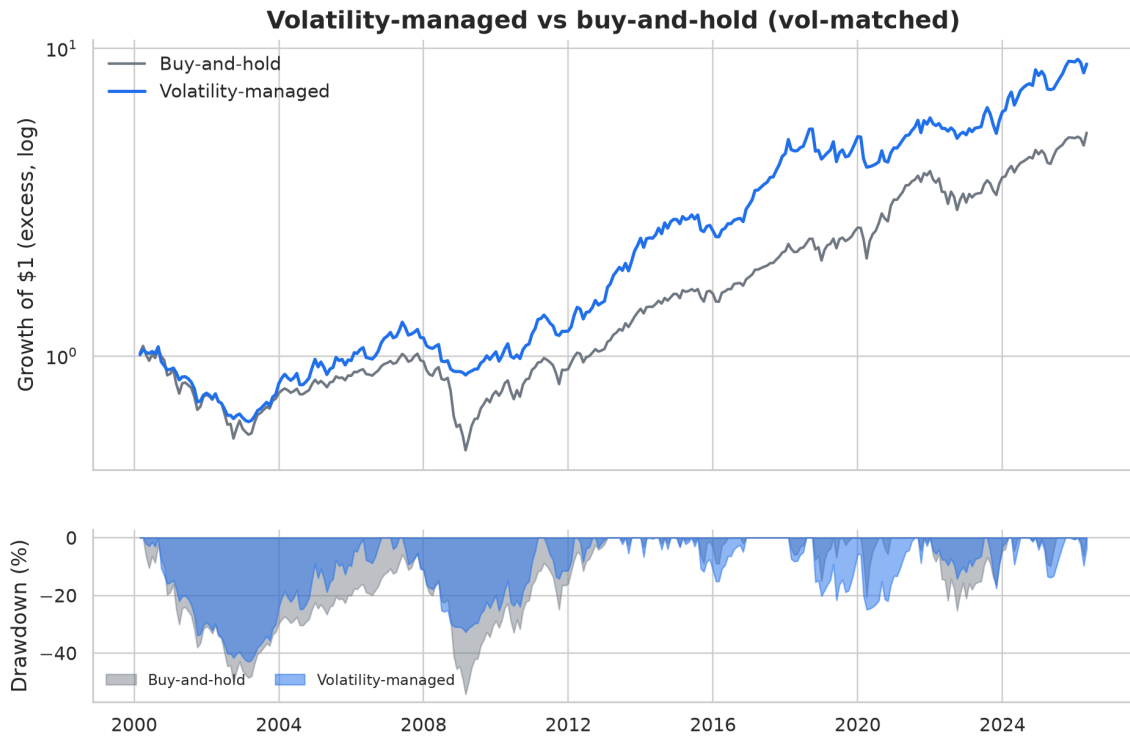


Figure 3: Time-varying exposure of the volatility-managed strategy.

Table 5: Headline performance: volatility-managed (naive forecast, 2× cap) vs buy-and-hold, vol-matched.

Metric	buy_and_hold	managed_headline
ann_return	0.076	0.096
cagr	0.066	0.087
ann_vol	0.158	0.158
sharpe	0.484	0.610
sortino	0.705	0.910
max_drawdown	-0.542	-0.429
calmar	0.121	0.202
skew	-0.479	-0.539
excess_kurtosis	0.789	0.777
n_months	315.000	315.000

**Figure 4:** Growth of \$1 in excess returns (log scale) and drawdowns, managed vs. buy-and-hold at matched full-sample volatility.**Table 6:** Managed-strategy performance by volatility forecaster (common sample, uncapped).

forecaster	ann_return	ann_vol	sharpe	max_drawdown	alpha_annual	alpha_tstat	appraisal_ratio
naive	0.090	0.151	0.597	-0.442	0.030	0.966	0.248
har	0.108	0.151	0.715	-0.355	0.026	1.263	0.307
garch	0.113	0.151	0.747	-0.327	0.035	1.561	0.374
ensemble	0.111	0.151	0.735	-0.337	0.031	1.433	0.348
buy_and_hold	0.099	0.151	0.655	-0.514	0.000	—	—

6 Robustness

Transaction costs. The strategy retrades as the leverage target moves. Charging a proportional cost on turnover (Table 7, Figure 5), the edge survives up to a break-even of roughly 39 bps per unit of turnover — comfortably above realistic costs for a liquid index exposure.

Leverage cap. Uncapped, the naive strategy occasionally takes large positions and inherits fat left tails; the literal Moreira and Muir (2017) estimator without a cap earns an alpha of 3.4% ($t = 1.22$). Capping exposure at 1.5–3 $\times$ and re-normalising volatility both controls the tails and improves risk-adjusted performance (Table 8), which is why the headline uses a 2 $\times$ cap.

Sub-periods. Table 9 splits the sample. The strategy helps most in turbulent, mean-reverting drawdowns and gives back some ground in sharp high-volatility rebounds — consistent with the regime decomposition below.

Table 7: Transaction-cost sensitivity of the headline strategy.

cost_bps	avg_turnover	ann_return	sharpe	alpha_annual	alpha_tstat
0	0.418	0.096	0.610	0.036	1.889
1	0.418	0.096	0.606	0.036	1.862
2	0.418	0.095	0.603	0.035	1.835
5	0.418	0.094	0.594	0.034	1.753
10	0.418	0.091	0.578	0.031	1.617
20	0.418	0.086	0.546	0.026	1.348

Table 8: Leverage-cap sensitivity (volatility re-normalised within each cap).

leverage_cap	mean_weight	max_weight	ann_return	ann_vol	sharpe	alpha_annual	alpha_tstat
1.000	1.000	1.000	0.076	0.158	0.484	0.000	–
1.500	1.191	1.500	0.096	0.158	0.612	0.031	1.850
2.000	1.259	2.000	0.096	0.158	0.610	0.036	1.889
3.000	1.244	3.000	0.094	0.158	0.593	0.040	1.764
inf	1.097	7.552	0.081	0.158	0.512	0.034	1.225

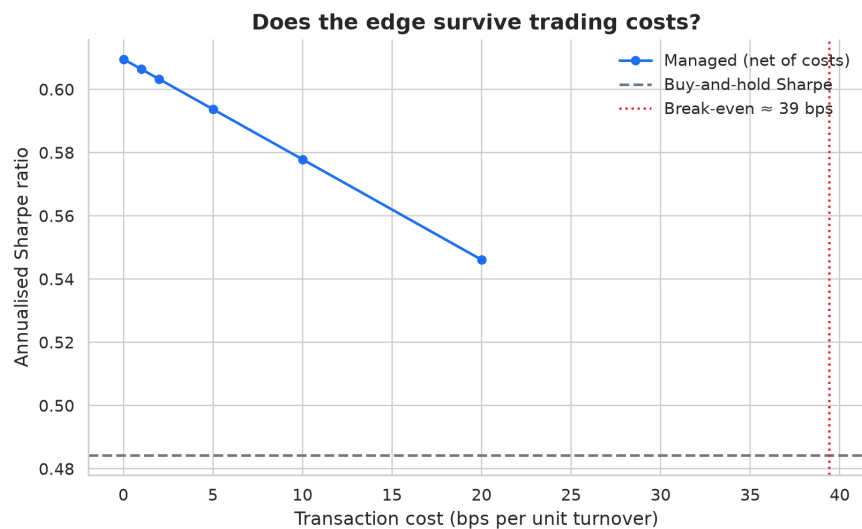


Figure 5: Net Sharpe ratio vs. transaction cost, with the break-even level.

Table 9: Sub-period analysis: managed vs buy-and-hold.

period	start	end	n	bh_return	bh_sharpe	mm_return	mm_sharpe	sharpe_gain
Dot-com bust	2000-02-29	2002-12-31	35	-0.166	-0.844	-0.150	-1.212	-0.368
Mid-2000s calm	2003-01-31	2007-06-30	54	0.124	1.398	0.160	1.188	-0.210
Global Financial Crisis	2007-07-31	2009-06-30	24	-0.205	-0.914	-0.147	-1.260	-0.346
Post-GFC bull	2009-07-31	2019-12-31	126	0.145	1.117	0.180	1.113	-0.004
COVID crash & rebound	2020-01-31	2020-12-31	12	0.247	0.893	-0.059	-0.263	-1.156
Post-COVID	2021-01-31	2026-04-30	64	0.106	0.680	0.132	0.789	0.109

7 Regime Analysis

Given only monthly returns and *no* crisis dates, the Gaussian HMM recovers the turbulent regimes — dot-com, the Global Financial Crisis, COVID-19 and the 2022 repricing all fall in the high-variance state (Figure 6, Table 10). This is unsupervised structure discovery, not curve-fitting to known events.

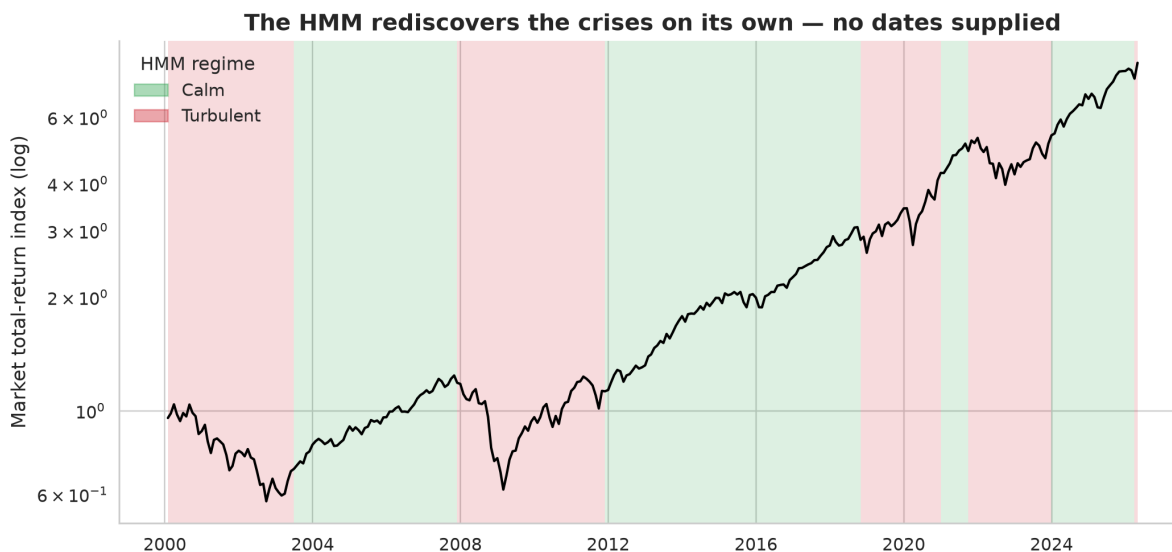


Figure 6: Market total-return index (log) shaded by the smoothed HMM regime. The model rediscovers the crises on its own.

Table 10: Estimated HMM regimes (states ordered by variance).

state	label	n_months	frequency	ann_mean	ann_vol	avg_duration
0	Calm	172	0.544	0.145	0.093	43.000
1	Turbulent	144	0.456	-0.010	0.208	28.800

Decomposing performance by regime (Table 11, Figure 7) reveals the mechanism. In calm regimes the strategy levers up and earns a higher return; in turbulent regimes it de-risks, cutting volatility well below buy-and-hold. The risk-adjusted gain therefore comes from two sources — harvesting the premium more aggressively when it is safe to do so, and protecting capital when it is not — partly offset by missing some of the most violent rebounds, which by construction occur in the turbulent state.

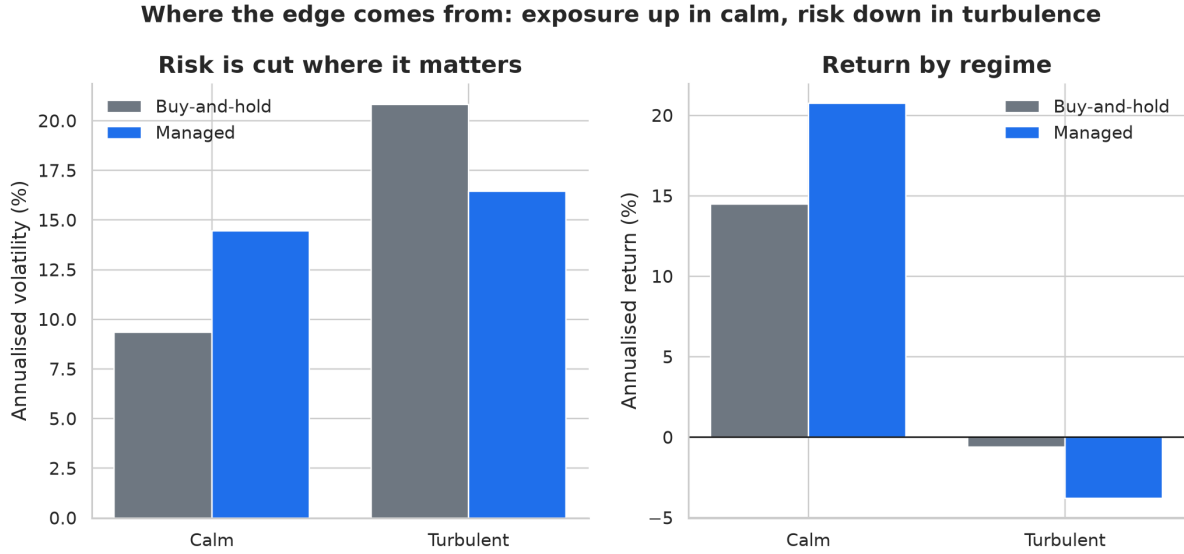


Figure 7: Volatility and return by regime, managed vs. buy-and-hold.

Table 11: Performance decomposition by (smoothed) regime.

regime	n_months	frequency	avg_exposure	mm_ann_return	bh_ann_return	mm_ann_vol	bh_ann_vol	monthly_out
Calm	172	0.546	1.588	0.207	0.145	0.144	0.093	0.000
Turbulent	143	0.454	0.864	-0.038	-0.006	0.164	0.208	-0.001

8 Multi-Asset Robustness and a Trend Overlay

Two questions test whether the result is a principle or an artifact: does volatility timing travel beyond the US market, and can we do better by diversifying and adding a complementary signal? We assemble a USD total-return panel of 11 assets across four classes — six equity regions (US, UK, Germany, Japan, France, emerging markets), Treasuries, gold, broad commodities, and Bitcoin — with excess returns over the same US risk-free rate. Their low cross-correlations (Figure 10) are the raw material for diversification.

Volatility timing alone does not travel uniformly (Table 12, Figure 8). It clearly helps US, UK and Japanese equities but hurts several others (Germany, France, parts of fixed income) — consistent with the more sceptical out-of-sample evidence of Cederburg et al. (2020). On its own it is largely an equity-index phenomenon, and honesty requires saying so.

The combination, however, is powerful. Adding a time-series-momentum overlay (Moskowitz et al., 2012) — long when the trailing twelve-month return is positive, short otherwise, with the same volatility-managed sizing — produces statistically significant alpha across all six equity regions and commodities. Combined into a diversified, inverse-volatility (risk-parity) portfolio, all vol-matched to the diversified buy-and-hold benchmark (Table 13, Figure 9), the full system lifts the Sharpe ratio from 0.39 to 0.72, cuts the maximum drawdown from -45.0% to -23.9%, and earns an annualised alpha of 8.0% ($t = 3.63$, Newey-West) — comfortably more significant than the single-market result, and 0.64 on a Sharpe basis even after 10 bps of trading costs. The lesson is the standard one of quantitative investing: the durable gains come from diversification and risk control, not from a single secret signal.

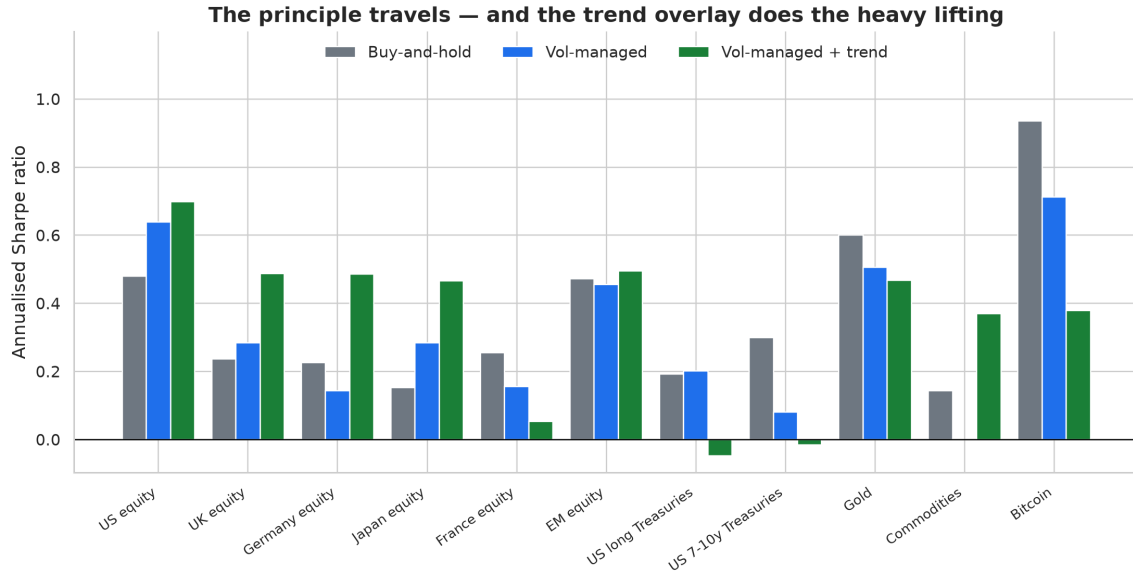


Figure 8: Sharpe ratios across the asset panel: buy-and-hold, volatility-managed, and volatility-managed with a trend overlay. Volatility timing alone is mixed; the trend overlay travels much more broadly.

Table 12: Volatility management and a trend overlay across assets (Sharpe ratios and managed-alpha t -statistics).

asset	class	n_months	BH SR	Managed SR	Mgd+Trend SR	Mgd alpha t	Trend alpha t
US equity	Equity	315	0.479	0.639	0.699	2.171	2.328
UK equity	Equity	315	0.236	0.284	0.487	0.781	2.320
Germany equity	Equity	315	0.226	0.143	0.486	-0.269	2.265
Japan equity	Equity	315	0.152	0.284	0.466	1.453	2.130
France equity	Equity	315	0.255	0.156	0.054	-0.347	0.168
EM equity	Equity	276	0.473	0.455	0.496	0.571	2.198
US long Treasuries	Bonds	284	0.193	0.201	-0.048	0.394	-0.455
US 7-10y Treasuries	Bonds	284	0.299	0.080	-0.015	-1.335	-0.508
Gold	Commodity	257	0.600	0.506	0.468	0.120	1.248
Commodities	Commodity	242	0.144	0.000	0.371	-0.753	1.546
Bitcoin	Crypto	139	0.936	0.713	0.378	0.004	0.557

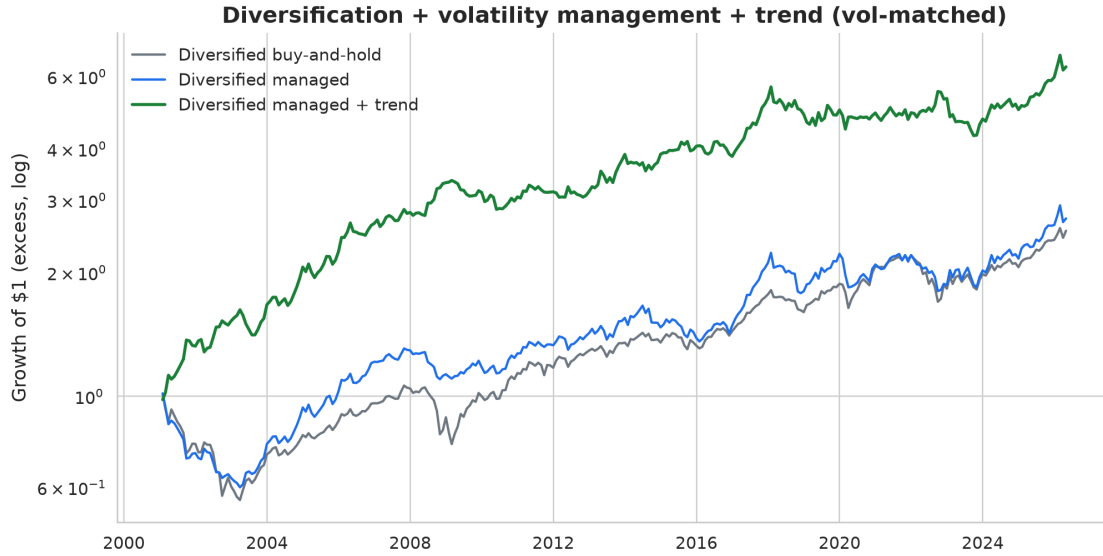


Figure 9: Diversified portfolios (excess growth of \$1, log scale), all vol-matched to the diversified buy-and-hold benchmark.

Table 13: Diversified portfolios vs the single-market strategy (all vol-matched to the diversified buy-and-hold).

strategy	ann_return	ann_vol	sharpe	max_drawdown	sortino	alpha_annual	alpha_tstat
US-only managed	0.096	0.158	0.610	-0.429	0.910	0.036	1.889
Diversified buy-and-hold	0.043	0.110	0.389	-0.450	0.545	—	—
Diversified managed	0.046	0.110	0.413	-0.409	0.595	0.010	0.796
Diversified managed + trend	0.079	0.110	0.719	-0.239	1.214	0.080	3.630

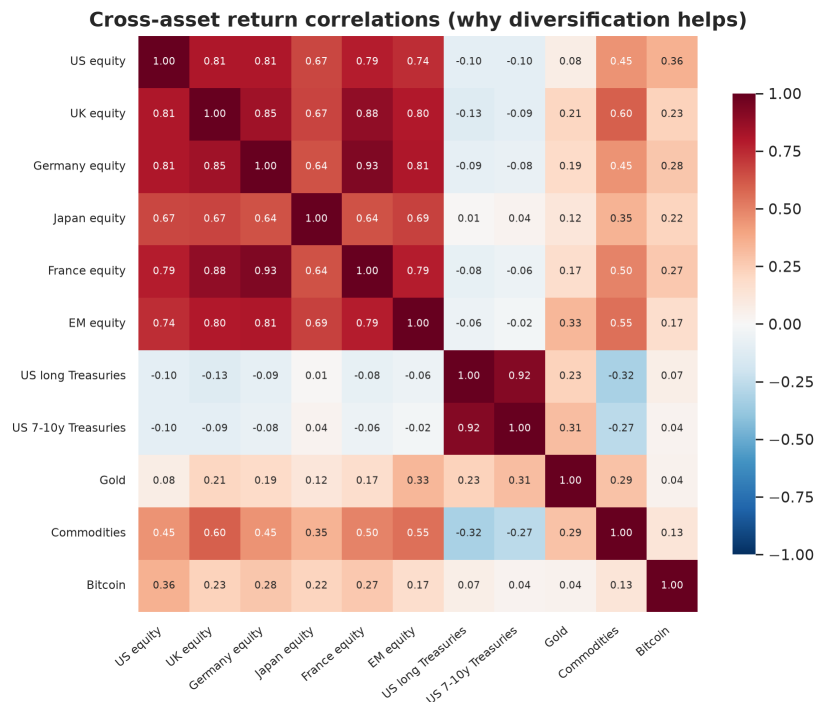


Figure 10: Monthly excess-return correlations across the asset panel — the source of the diversification benefit.

9 Discussion and Limitations

The Deflated Sharpe Ratio of 0.999 (computed on the common evaluation sample across the three forecasters tried) indicates the managed Sharpe ratio is unlikely to be a multiple-testing artifact, even after correcting for non-normality, sample length and the number of configurations searched. Several caveats remain. (i) A backtest is not live trading: we model a turnover cost but ignore market impact, financing spreads and the practical frictions of monthly rebalancing leverage. (ii) Realized variance uses daily data; intraday data would sharpen both the HAR model and the volatility proxy. (iii) Results are sample-dependent: the 2000–2026 window combines a secular decline in interest rates with two unusually fast crashes, and the strategy’s value is concentrated in turbulent regimes. (iv) Capacity is finite and the leverage assumption is idealised. (v) The trend overlay is itself a well-known, widely-traded signal, and the diversified result depends on the chosen asset panel and on going short and using leverage in practice. *This is a research study and explicitly not investment advice.*

10 Conclusion

You cannot predict the price, but you can predict the risk — and that is enough to improve risk-adjusted returns. Replicating Moreira and Muir (2017) on the US market factor with strict out-of-sample discipline raises the Sharpe ratio from 0.48 to 0.61 (0.75 with a GARCH forecaster on the common sample), cuts drawdowns, survives realistic costs and a deflated-Sharpe correction, and is explained — not merely reported — by an unsupervised regime model that rediscovers the crises and shows the strategy earning its keep by sizing risk to the regime. Pushed across 11 assets and paired with a trend overlay in a diversified portfolio, the same discipline reaches a Sharpe ratio of 0.72 with half the drawdown and an alpha significant at $t = 3.63$

— a reminder that in investing the reliable edges are diversification and risk management, not the prediction of prices.

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A Plain-Language Summary (for non-specialists)

What problem are we solving? Nobody can reliably say whether the stock market will go up or down tomorrow — that part really is close to a coin flip. But there is something we *can* see coming: how *shaky* the market is about to be. Calm weeks tend to be followed by calm weeks, and stormy weeks by stormy weeks. Risk clusters together in time; direction does not.

The idea, in one picture. Think of driving a car. You cannot predict every bend in the road ahead — that is the *price*. But you can feel when the road turns icy — that is the *volatility*. A good driver does not try to guess the bends; they simply **slow down on the ice and speed up when the road is clear**. They arrive safer, and barely any later. This strategy does exactly that with money: it invests *more* when markets are calm and *less* when markets are turbulent, without ever trying to guess the direction.

Does it work? Yes, in a careful and honest sense. At the *same* level of risk as simply holding the market, it earned a little more and — more importantly — it fell much less during crashes (its worst loss was about −43% instead of −54%). The ride is smoother. We stress-tested it hard (realistic trading costs, limits on borrowing, and a statistic that specifically punishes results that could be down to luck), and it held up.

But is it magic? No. It is a respected, well-known professional technique, not a money machine. Three honest caveats:

- It does *not* predict prices, so it will not make anyone rich quickly. It improves the *quality* of returns (less stress, smaller crashes) more than it boosts their raw size.
- A test on past data is not a promise about the future. The advantage was real but modest over this period, and it depends on the era.
- Because the idea is well known, many investors already use it, which can slowly wear the advantage down.

Where does it go from here? The biggest, most reliable gains in investing come from *diversification* and *risk control*, not from a secret formula. We took exactly those steps: running the same risk-sizing idea across 11 markets and asset classes at once (stocks around the world, bonds, gold, commodities, even Bitcoin) and adding *trend-following* — simply leaning with the wind, going with what has been rising and away from what has been falling. Together, in a diversified portfolio, this nearly doubled the risk-adjusted performance and roughly halved the worst loss. Importantly, even this makes the system *smoother and more robust* — it is not a guaranteed path to riches, and it relies on being able to short and to use leverage.

The one-sentence takeaway. You cannot predict the price, but you can manage the risk — and managing risk well is, over the long run, one of the few things in investing that reliably pays off.